



SPONTOM ENTERPRISE PRIVATE LIMITED

Build with AI (Google) • Track 2

Vayu Drishti

**From a citizen's photo
to an officer's action.**

Hyper-local air-quality intelligence that fuses citizen, sensor, satellite and weather evidence for the officer who can act.

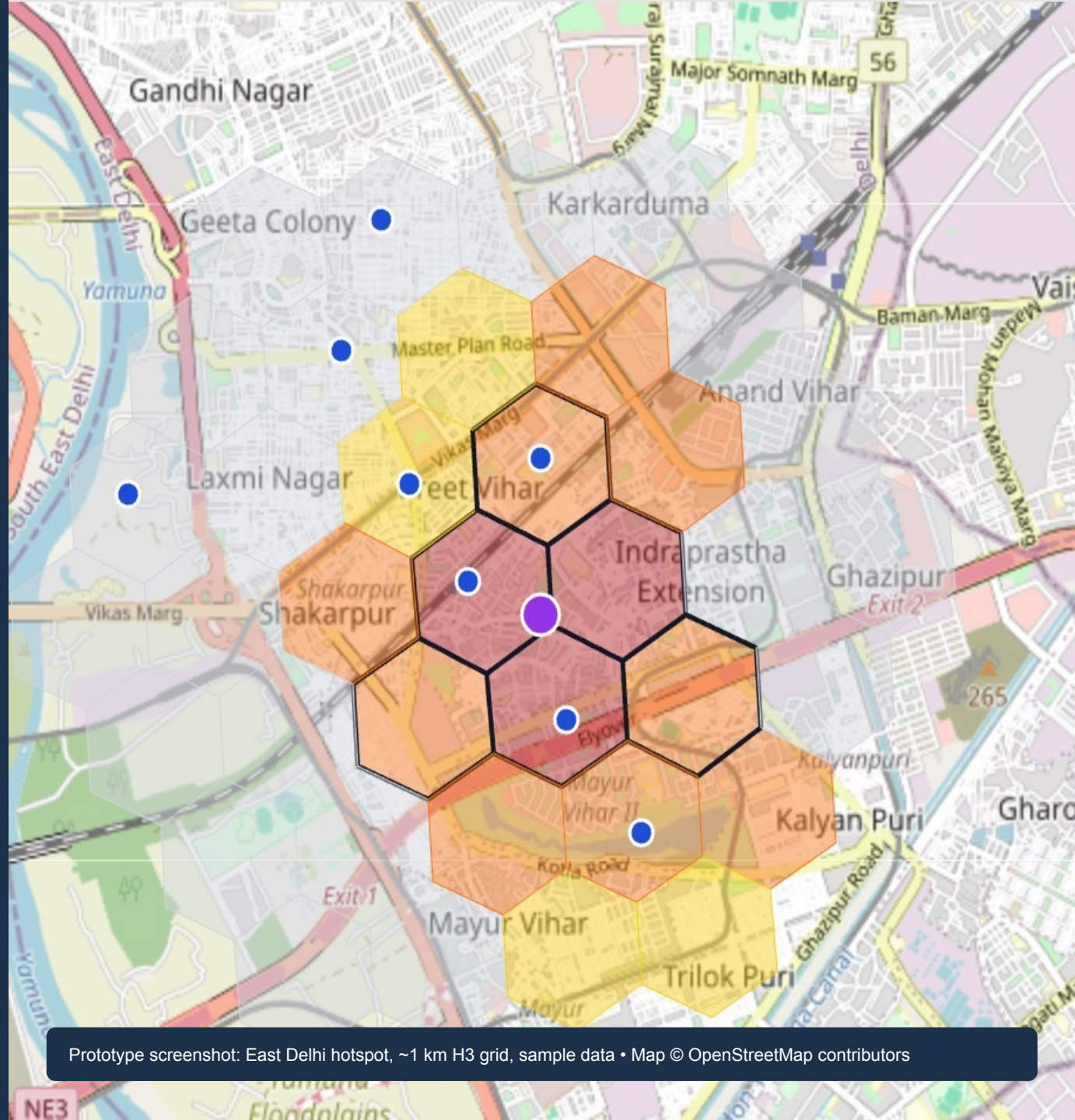
GEMINI-POWERED

OFFICER-APPROVED

FEDERATED

A working prototype by Spontom Enterprise Private Limited

DPIIT Recognised • Visakhapatnam • September 2026



Prototype screenshot: East Delhi hotspot, ~1 km H3 grid, sample data • Map © OpenStreetMap contributors

The station reads the city. The smoke is on one street.

India monitors macro-level air quality well, but misses hyper-local events such as night-time industrial smoke. [1]

One station average hides
a street-level event.

4.9 km

to nearest station

249

µg/m³ local PM2.5

66

score · alert at 70

Near an East Delhi industrial estate the cell reads 249 µg/m³ against 112 at the station. It scores 66, so nobody is alerted.

THE LOCAL BLIND SPOT

SIGNALS

Stations, sensors,
photos, satellite



SILOS

Held in separate
systems



?

No fused, verified
evidence



REACT

Action after the peak

FOCUSED PROBLEM

Officers lack verified, located evidence of hyper-local events in time to inspect.

Consequence

Sources go undetected, inspections are poorly targeted and smoke crosses state lines unannounced.

[1] VayuDrishhti PRD §2.1. [2] CPCB National AQI. Numbers: VayuDrishhti demo run on labelled sample data (ITO sample station, simulated sensors), not real readings.

VayuDrishti turns scattered signals into one inspection.

Gemini works only on structured evidence; every consequential step waits for an officer.

01

SEE

Citizen photo, sensors, satellite and weather fused on a ~1 km grid.

02

SCORE + BRIEF

Transparent Hotspot Confidence; Gemini drafts likely source and inspection.

03

HUMAN DECISION

Officer approves actions; advisories go out in hi/pa/en with voice.

AI ACTION BRIEF • GEMINI 2.5 FLASH

DRAFT • NEEDS APPROVAL

Sangrur–Patiala belt, Punjab • 95/100 • severe

1

Likely source: crop residue burning (confidence 0.95).

2

Evidence: PM2.5 226.4 µg/m³ (indicative) • 39 fires upwind.

3

Recommended: dispatch cluster nodal officer / field team.

4

Outlook: ~287 µg/m³ (Severe) at +48 h.

⚠

Uncertainty stated: nearest official station 53.1 km away.

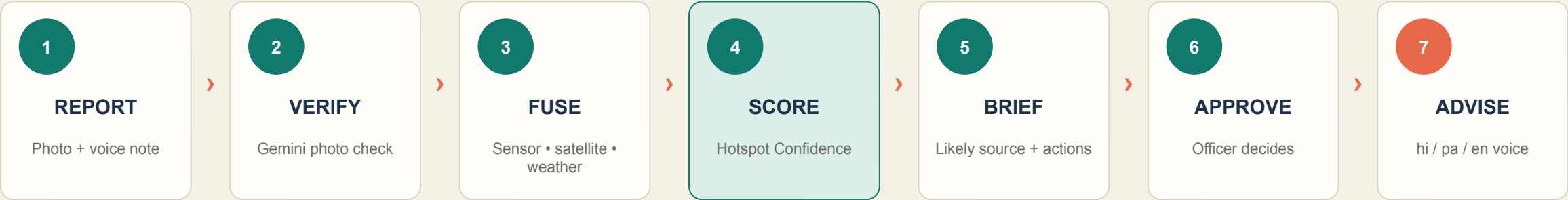
Prompt action_brief_v2 • numbers checked vs evidence

APPROVE ACTIONS

Real Gemini output read from the deployed API (30 Sep 2026), generated on labelled sample sensor/satellite data; wording condensed.

From one photo to an approved action in seven steps.

Machines gather and rank evidence; people approve anything that leaves the system.



TIER 0 • AUTOMATIC

Photo check, fusion and scoring run in seconds. Nothing public goes out unapproved.

TIER 1 • OFFICER

Scores at or above the threshold (Delhi 70, Maharashtra 65) reach the jurisdiction's inbox.

TIER 2 • CROSS-STATE

An upwind hotspot (Punjab → Delhi NCR) becomes an officer-approved cross-boundary alert.

CALIBRATION ASSUMPTION **Weights 35/25/25/15 and thresholds are configured defaults to tune against pilot inspection outcomes.**

Hotspot Confidence = sensor anomaly 35% • satellite 25% • citizen reports 25% • weather plausibility 15% (services/api/app/hotspots/confidence.py).

Design for the officer first; connect the people around her.

Two app roles (PRD §7): Environmental Officer (primary) and Citizen Reporter (secondary).

HOTSPOT CONFIDENCE · 6 CELL(S)

83 / 100 High alert threshold 70

Sensor anomaly × 35%

100 → +35.0

Local PM2.5 249 ug/m3 vs 112 expected (+123%); PM10 431 vs 230

Satellite signal × 25%

64 → +16.0

NO2 +38% vs 30-day median; 0 fire detections upwind (100 km); AOD 1.13 (strongest signal counts)

Citizen reports × 25%

69 → +17.2

1 verified report in/near cell

Weather plausibility × 15%

99 → +14.8

Wind 1.1 m/s, boundary layer 220 m; candidate source upwind or in cell

RECOMMENDED INSPECTION (APPROVE TO RECORD)

☒ high Night-time inspection of industrial units upwind of the affected cells DL-INSPECT-IND-NIGHT

☐ medium Check stack emission monitoring / online CEMS data of nearby units DL-STACK-MONITOR

☐ medium Coordinate with traffic police to decongest the corridor at peak hours DL-TRAFFIC-DECONGEST

☐ medium Mechanised road sweeping and water sprinkling in affected wards DL-DUST-SPRINKLE

Acknowledge & approve selected actions

ENVIRONMENTAL OFFICER · PRIMARY USER

One alert: score, evidence and an inspection to approve

01

CITIZEN

Reporter (secondary)

Reports by photo or voice in Hindi, Punjabi or English; sees status and hears advisories.

02

FIELD TEAM

Acts on approval

Gets a located, prioritised inspection instead of a city-wide average.

03

NEIGHBOUR SPCB

Cross-boundary

Receives approved alerts for upwind smoke, e.g. Punjab → Delhi NCR.

04

CAQM / STATE

Oversight

Sees audit trails and when a GRAP-stage review is recommended.

CONSTRAINT → DESIGN RESPONSE

FEW STAFF

ranked, located briefs

LANGUAGE

hi / pa / en + voice

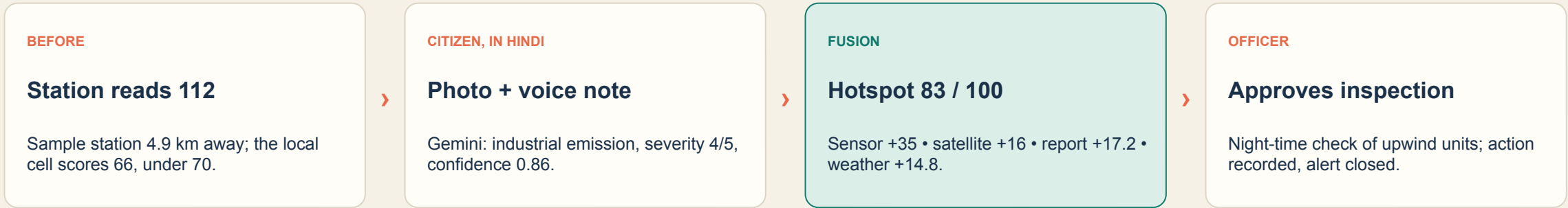
PATCHY DATA

freshness + labels

Screenshot: officer dashboard, local demo run on labelled sample data. Stakeholders beyond the two roles are reached through approved alerts.

Scenario A: night-time smoke in East Delhi.

One Hindi voice report pushes a hidden hotspot over the threshold (PRD §43–44).



OUTCOME

One photo → one approved inspection.

24 h outlook: ~201 µg/m³ (185–218), Very Poor.

+

Voice advisory in hi / pa after approval

✓

Audit trail for every step

↗

Cross-boundary alert if upwind

WHY OFFICERS RETURN

Prioritised, located work where every number is traceable to a source and a timestamp.

Local demo run, 30 Sep 2026, on labelled sample/simulated data; routed to the DPCC demo inbox. Live Gemini values vary by a point or two.

Not another AQI dashboard. A bridge from evidence to action.

Existing systems do essential work; VayuDrishti focuses on the local event nobody has verified yet.

APPROACH	WHAT IT CONTRIBUTES	REMAINING BOUNDARY / GAP
CPCB stations + National AQI	Regulatory-grade readings and a public AQI scale [2]	Sparse stations can miss a street-level event [1]
SAFAR / IITM forecasts	City-scale AQI and forecasts for major metros [3]	Forecasts, not a verified-report-to-inspection workflow
GRAP (CAQM)	Staged, region-wide measures for Delhi NCR [4]	Region-wide; does not locate a specific source
Complaint apps / portals	Citizens can flag visible pollution	Reports hard to verify; not fused with sensors or satellite [1]
VayuDrishti	Verified photo + sensor + satellite + weather → brief → approval	Complements the above; accuracy unvalidated beyond sample data

DISTINCTIVE CAPABILITY

A verified citizen photo becomes an officer-approved, cross-state-aware inspection.

[1] PRD §2.1. [2] CPCB National AQI. [3] SAFAR, IITM Pune (MoES). [4] Commission for Air Quality Management.

Bounded AI on a federated evidence grid.

Gemini interprets structured evidence; it never invents measurements.

01

CITIZEN CHANNEL

Next.js app • photo + voice (Speech-to-Text) • hi/pa/en • Maps geocoding

02

STATE ADAPTERS

3 vendor feeds → canonical schema • κ-Köhler humidity calibration [6]

03

FUSION GRID

H3 res 8 (~1 km) • Earth Engine layers [5] • 24–72 h shared forecast model

04

DECISION LAYER

Gemini on the evidence bundle • action rules only • numeric grounding check

05

OFFICER + DELIVERY

Inbox • approve / record / close • Translation + TTS • BigQuery audit

GOOGLE AI INTEGRATION MAP

Gemini 2.5 Flash (Vertex AI): photo, brief, advisory [7]

Translation • Text-to-Speech • Speech-to-Text

Maps • BigQuery • Cloud Storage • Cloud Run

WIRED

 Earth Engine + Vertex Model Registry: on sample until access

REAL-VS-SAMPLE RULE

Live

 Gemini, Translation, TTS/STT, Maps, BigQuery, GCS

Sample

 EE layers, CPCB station, low-cost sensors: labelled

Freshness

 Each signal shows source + age; header badge: live vs sample

[5] Earth Engine catalogue (Sentinel-5P NO2, FIRMS). [6] Crilley et al. (2018), AMT. [7] Gemini 2.5 Flash on Vertex AI.

Likely source, not a verdict. Humans decide.

The platform guides inspections; it does not enforce, penalise or broadcast on its own (PRD §6).

HUMAN APPROVAL

NO AUTO-BROADCAST • NO ENFORCEMENT

Alerts

Start as pending review; the officer approves, records the action and closes.

Transfers

Cross-boundary alerts and public advisories each need explicit approval.

DO NOT COLLECT / CLAIM

- Reporter identity beyond a pseudonym
- Precise reporter locations in public views
- “Polluter” labels: only a likely source

SAFETY BY DESIGN

GROUNDING

Gemini sees only the evidence bundle; never invents measurements.

NUMERIC CHECK

Numbers absent from the evidence are flagged as warnings.

RULE BOUND

Actions outside the jurisdiction’s rules are dropped.

PROVENANCE

Model, version and prompt version stored on every AI record.

HONEST DATA

Low-cost sensors labelled indicative; sample data labelled.

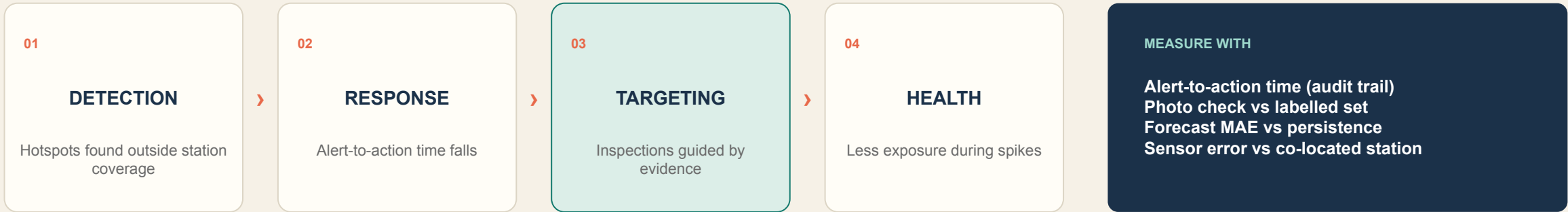
PHOTO CHECK

Irrelevant or poor photos are rejected and kept out of the score.

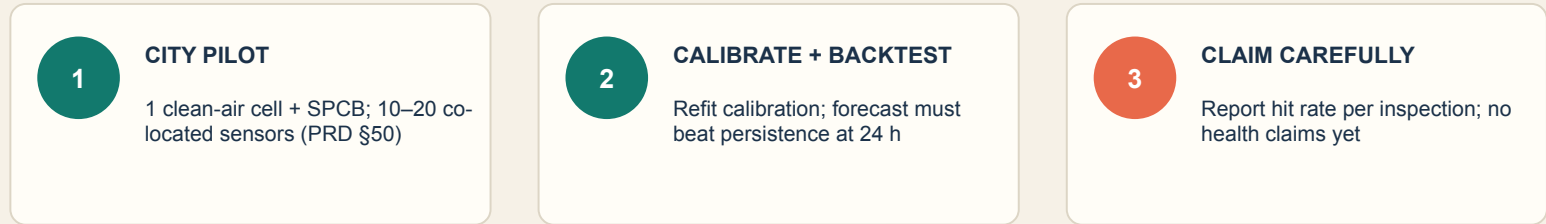
PRD §6, §35, §52. Roadmap: face and number-plate blurring before any public display; role enforcement and login.

Earn the right to scale, one clean-air cell at a time.

The first proof is not app activity. It is a verified hotspot that led to a timely inspection.



VALIDATION GATES



BUDGET ROUTE

NCAP city clean-air action-plan funding (MoEFCC), tracked on PRANA [8]; pilot fit to confirm with the SPCB.

OWNERSHIP + SCALE

Owner	State Pollution Control Board / city clean-air cell
Channel	SPCB field teams; neighbour boards via adapters
Scale	New state = JSON adapter + language pack

Current model accuracy uses simulated histories (e.g. Delhi 24 h MAE 13.8 vs 16.7 persistence) and demonstrates the method only.

Working today, end to end.

The header badge in both apps shows which integrations are live and which run on sample data.

आपकी रिपोर्ट RP-4260D1A5 verified

AI photo check: demo-fixture · photo_verification_v1 · Demo mode: stored fixture result for sample photo industrial_chimney_smoke.jpg

Pollution event detected · **industrial emission** · severity 4/5 · confidence 86%

dense smoke plume from a tall stack continuous emission industrial chimney

A dense smoke plume is rising from an industrial chimney.

धुंधलवादा। आपकी रिपोर्ट अभी गई और स्वस्थता राक्षस में जोड़ी गई।
Hotspot Confidence here: 66 → **83** / 100 (alert threshold 70)
✓ समीक्षा के लिए पर्यावरण अधिकारी को भेजा गया (Delhi Pollution Control Committee (demo inbox))

स्वास्थ्य सलाह

आपके क्षेत्र के लिए अभी कोई स्वीकृत सलाह नहीं है।

मेरी रिपोर्टें

Citizen app: Hindi report, AI verdict, 66 → 83

VayuDrishhti Environmental Officer

Delhi Pollution Control Committee (demo inbox) · East Delhi industrial + traffic corridor

Delhi NCR (DL-DPCA) · officer: dpc

Active images: 1

Active monitoring events: 1

Active incidents / alerts: 0 / 0

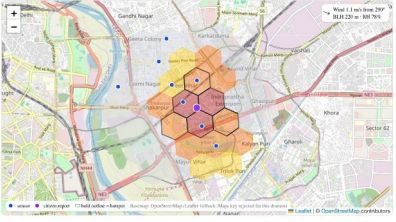
Checked reports: 1 (1 verified)

Checked monitoring in real: 0 / 0

Officer's rate: 112 pg / 100 index

Hyper-local grid (RS as 8, ~1 km cells)

Color: Hotspot Confidence · Forces: new · sample / rendered data



Alert inbox

Blocked to this jurisdiction. Every action needs you

AL-DS-5653BF

sample data

AL-DS-5653BF

Delhi Pollution Control Committee (demo inbox) · sent 0 min ago · hotspot RS-DS-402637

AL-DS-5653BF (demo) from your location

Likely industrial emission affecting a grid cell in East Delhi industrial + traffic corridor, Delhi NCR, Haryana

83

Officer dashboard: hotspot map + alert inbox

EVALUATION ALIGNMENT

25%

AI / TECHNICAL EXECUTION

Gemini multimodal + schema-validated briefs, numeric grounding, provenance

20%

PROBLEM-SOLUTION FIT

Finds events stations miss; routes evidence to the officer who can act

20%

DEPTH & REACH ACROSS INDIA

3 state adapters, canonical schema, hi/pa/en voice, cross-state alert

20%

DEPLOYABILITY & SCALABILITY

Cloud Run + BigQuery + GCS; a new state is a JSON adapter

15%

IMPACT POTENTIAL

India's most polluted corridors; approved health advisories

TRY IT

Citizen app

<https://air-quality-network-citizen-web-847963771142.asia-south1.run.app>

Officer dashboard

<https://air-quality-network-officer-dashboard-847963771142.asia-south1.run.app>

API (/docs)

<https://air-quality-network-api-847963771142.asia-south1.run.app/docs>

Source

<https://github.com/ipavanreddy/air-quality-network>

Screenshots: local run on labelled sample data, 30 Sep 2026. Map © OpenStreetMap contributors; sample photo CC BY-SA (THIRD_PARTY.md).

Problem evidence & public systems

Numbered references correspond to citations in the core proposal.

[1]

VayuDrishti PRD (Track 2 brief)

docs/PRD.md §2 (problem), §43–44 (scenarios), §50 (scale). Challenge framing and demo design.

LOCAL EVIDENCE

[2]

CPCB: National Air Quality Index

Central Pollution Control Board. AQI categories and the national real-time monitoring network.

[Official source ↗](#)

[3]

SAFAR, IITM Pune (MoES)

System of Air Quality and Weather Forecasting and Research: AQI and forecasts for Delhi, Pune, Mumbai, Ahmedabad.

[Official source ↗](#)

[4]

Commission for Air Quality Management

Statutory body for Delhi NCR and adjoining areas; administers the Graded Response Action Plan (GRAP).

[Official source ↗](#)

Data, methods & funding route

Datasets and models are also listed with licences in THIRD_PARTY.md.

[5]

Earth Engine data catalogue

Sentinel-5P TROPOMI NO2 (COPERNICUS/S5P/OFFL/L3_NO2); NASA FIRMS, MAIAC AOD, ERA5-Land and GFS are also catalogued.

[Dataset ↗](#)

[6]

Crilley et al. (2018)

“Evaluation of a low-cost optical particle counter (Alphasense OPC-N2) for ambient air monitoring.” Atmos. Meas. Tech. 11, 709–720. Basis of the κ -Köhler humidity correction.

[Paper ↗](#)

[7]

Gemini 2.5 Flash on Vertex AI

Google Cloud model documentation for the model used for photo verification, action briefs and advisories.

[Model docs ↗](#)

[8]

PRANA: NCAP portal (CPCB)

Portal for Regulation of Air-pollution in Non-Attainment cities: tracks city action plans under the National Clean Air Programme.

[Official source ↗](#)